

Marketplace Logistics Cost Optimization

Executive Summary | Submitted by: Arya Manjardekar | April 2026

Overview

This report analyzes **100 marketplace orders from January 2025** across 7 e-commerce platforms shipped from 3 warehouses (Mumbai, Bangalore, Delhi). Total logistics cost is **Rs. 25,00,333**. The analysis identifies **Rs. 1.54L - Rs. 2.40L in monthly savings opportunities** through targeted operational improvements — no new infrastructure required.

Cost Breakdown — 100 Orders

Cost Component	Amount (Rs.)	Share
Transportation	11,72,717	47%
Stickering (Amazon only)	11,84,388	47%
Appointment Penalties	1,43,228	6%
Total Logistics Cost	25,00,333	100%

Key Findings

- **Transportation dominated by 2 platforms:** Amazon (47.4%) and Flipkart (36.9%) account for 84.3% of spend. Amazon pays Rs. 13.87/kg vs Flipkart's Rs. 11.23/kg — a 23% rate premium.
- **Appointment adherence is critically low:** Overall 45% vs industry standard of 70%. Blinkit worst at 27.3%, Amazon best at 53.3%.
- **Bangalore underutilized despite being cheapest:** At Rs. 2.513/unit vs Mumbai's Rs. 2.969/unit (18% cheaper), Bangalore handles only 28.3% of volume. Rebalancing saves Rs. 28,778/month.
- **Stickering manpower swings 7.5x in a month:** Peak 264 workers (Jan 19) vs minimum 35 (Jan 25). Recommended: 228 permanent + 36 on-call temp roster.

Savings Opportunity Summary

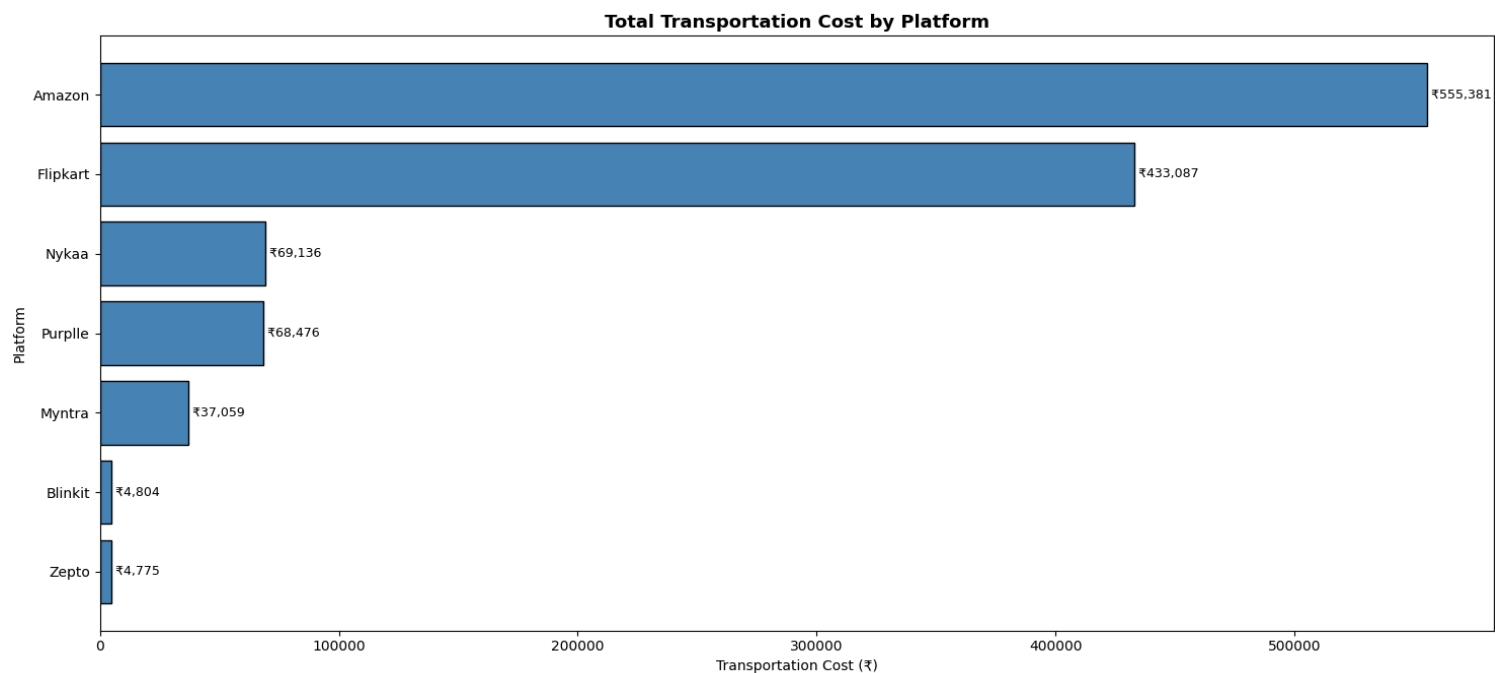
Action	Est. Monthly Saving
Renegotiate Amazon freight rate to match Flipkart benchmark	Rs. 75K - Rs. 1.05L
Reduce penalties via buffer scheduling + Flipkart prioritization	Rs. 30K - Rs. 45K
Redirect Mumbai volume to Bangalore (geographic routing)	Rs. 29K - Rs. 50K
Right-size manpower: 228 permanent + 36 temp on-call	Rs. 20K - Rs. 40K
Total Monthly Savings Opportunity	Rs. 1.54L - Rs. 2.40L

AI / Automation Roadmap

- **Penalty prediction model** (High) — ML model flagging high-risk orders before dispatch using platform, warehouse, transit time, day-of-week.
- **Warehouse routing engine** (High) — Rule-based system assigning optimal warehouse per order based on destination pincode + load + cost.
- **7-day manpower forecast** (Medium) — Automated weekly staffing plan from appointment data. Eliminates reactive crisis management.
- **Rate benchmarking dashboard** (Medium) — Live tracking of Rs./kg per platform-warehouse with auto-alerts for rates 15%+ above rolling average.

Excel Q & A

Q1. Calculate the total transportation Cost for each platform and identify opportunities to optimize cost.



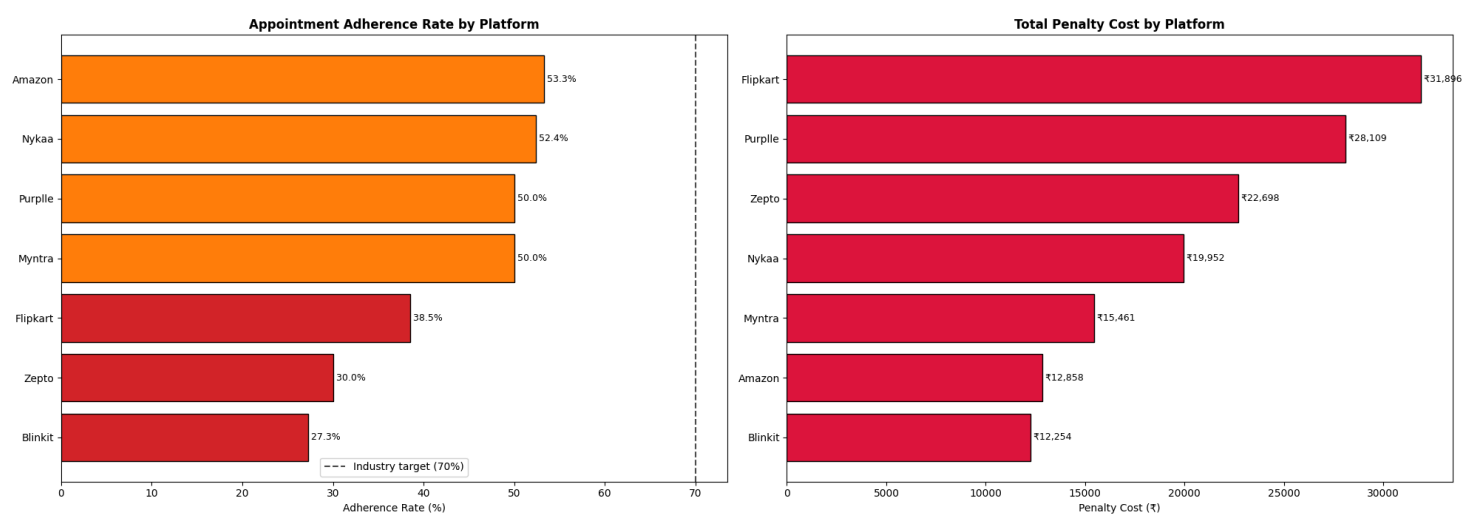
Recommendations (3 Actions)

Recommendation 1: Renegotiate Amazon freight contracts Benchmark Amazon's ₹13.87/kg rate against Flipkart's ₹11.23/kg on similar routes. Even bringing Amazon down to ₹12/kg saves: 39,884 kg × ₹1.87 = **₹74,583 per month** at current volume.

Recommendation 2: Consolidate small orders into FTL shipments Multiple small Nykaa/Purplle orders going to the same city on the same day should be combined into one Full Truck Load. FTL rates are typically 20–30% cheaper than LTL (Less Than Truck Load).

Recommendation 3: Build a rate benchmarking tracker Track ₹/kg per platform every week. Flag any rate 15%+ above the rolling average for renegotiation. This turns freight pricing from a one-time decision into a continuous optimization.

Q2. Calculate the total Stickering cost for each platform and analyze appointment adherence rates. Provide recommendations to reduce penalties



Written Findings -

Finding 1: No platform meets the 70% industry adherence standard Best performer is Amazon at 53.3%. The worst is Blinkit at 27.3%. The company-wide adherence average is **45%** — meaning more than half of all appointments are missed. This is a systemic operations problem, not a platform-specific one.

Finding 2: Flipkart drives the highest penalty cost despite mid-level adherence Flipkart's 38.5% adherence generates ₹31,896 in penalties — the highest of any platform. Flipkart likely has higher per-miss penalty rates, making it the highest priority to fix.

Finding 3: Quick commerce platforms (Blinkit, Zepto) have both the worst adherence AND high penalties. These platforms have the tightest delivery

windows (same-day or next-day). Missing their slots is structurally harder to avoid — but also structurally more expensive when missed.

Recommendation 1: Add 1-day buffer to all appointment scheduling Most missed appointments happen because dispatch is scheduled for the exact minimum transit time with zero buffer. Adding just 1 day of buffer to appointment booking would dramatically improve adherence — especially for Flipkart and Zepto.

Recommendation 2: Build a penalty risk prediction model Use historical data — platform, warehouse, transit time, order weight, day of week — to predict which orders are likely to miss appointments before they're dispatched. Flag high-risk orders for priority handling. This is where AI adds direct business value.

Recommendation 3: Prioritize Flipkart appointments operationally Since Flipkart generates the most penalty cost per miss, warehouse teams should treat Flipkart appointment orders as highest priority for packing and dispatch. A simple rule: Flipkart orders get packed first, every day.

Q3. Calculate the order Volume handled by each warehouse and propose a redistribution plan to balance warehouse load and minimize costs.

Written Findings -

Finding 1: Load is imbalanced — Delhi overloaded, Bangalore underutilized Ideal split is 33.3% each. Delhi handles 38% of orders, Bangalore only 28.3%. Bangalore has spare capacity that isn't being used.

Finding 2: Mumbai is the most expensive warehouse At ₹2.969/unit, Mumbai costs 18% more than Bangalore (₹2.513) and 14% more than Delhi (₹2.606). Yet Mumbai handles 34% of all units — a significant cost inefficiency.

Finding 3: Shifting volume from Mumbai to Bangalore saves money immediately If 20% of Mumbai's units (63,110 units) were redirected to Bangalore: saving = $63,110 \times (\text{₹}2.969 - \text{₹}2.513) = \text{₹}28,778$ per month at current volume. No new infrastructure needed — just routing rules.

Recommendations

Recommendation 1: Redirect platform orders geographically

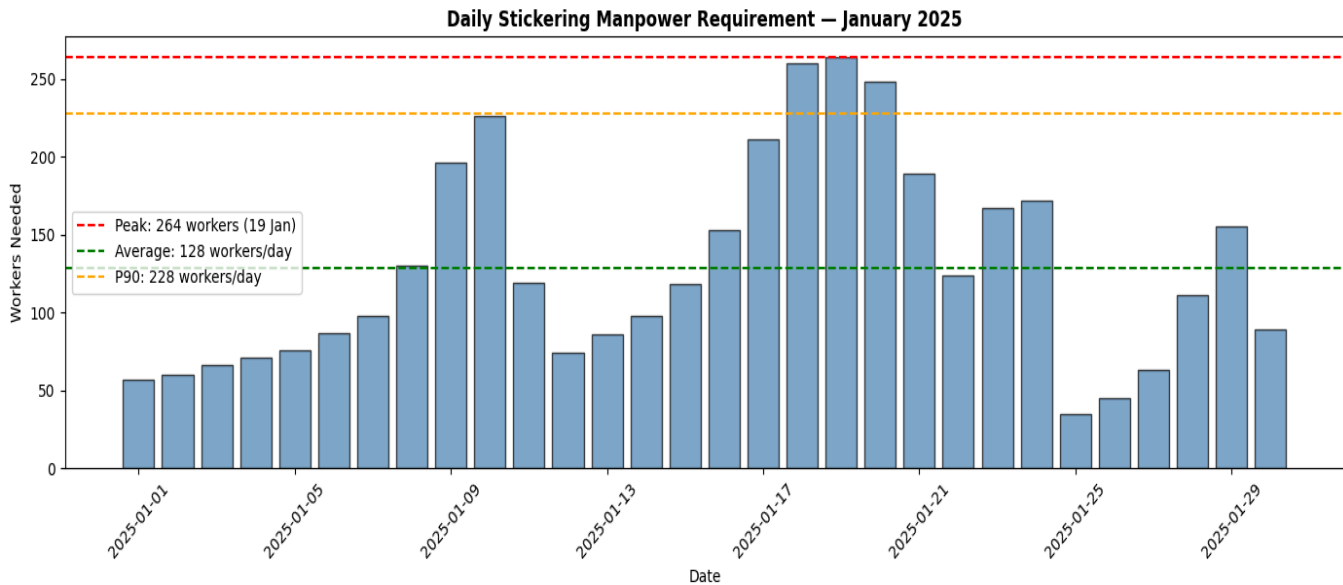
- Northern platform demand (Delhi-NCR consumers) → Delhi warehouse
- Southern/Western demand → Bangalore warehouse
- Mumbai warehouse for Maharashtra-specific demand only This naturally reduces Mumbai's load and increases Bangalore utilization.

Recommendation 2: Build warehouse routing rules into the order system

For each new order, automatically assign the warehouse based on: destination pincode + current warehouse load + cost per unit. A simple rule engine — no ML needed — can save ₹20K–₹30K/month.

Recommendation 3: Cap Mumbai warehouse at 30% of total volume Set an operational rule: Mumbai handles maximum 30% of total monthly orders. Overflow automatically routes to Bangalore. This is an easy policy change with immediate cost impact.

Q.4 Plan the Stickers manpower headcount for each order requiring stickering to ensure that the appointment timeline can be met without delay. Assume that today is 01-Jan-2025.



Written Findings

Finding 1: Peak demand (264 workers) is 7.5× the minimum demand (35 workers) Appointment dates are heavily clustered — multiple large Amazon orders have deadlines falling on the same week (Jan 17–19). This creates an artificial surge that makes permanent headcount planning extremely difficult.

Finding 2: Three distinct demand spikes exist in January Jan 9–10, Jan 17–19, and Jan 29–30. Each spike is driven by 2–3 large orders with coinciding deadlines. Spreading these appointment dates by just 3–4 days each would dramatically flatten the curve.

Finding 3: Average daily need (128 workers) is less than half the peak (264) Hiring to the peak wastes money on quiet days. Hiring to the average means missing appointments on spike days and incurring penalties.

Recommendations

Recommendation 1: Hire permanent staff to P90 (228 workers) This covers 90% of all days without temp workers. Only the top 10% of demand days (roughly 3 days/month) need supplemental staffing. Cost-efficient and operationally safe.

Recommendation 2: Maintain a temp worker roster of 36 people 264 (peak) – 228 (P90) = 36 temp workers needed for spike days. Keep a pre-vetted roster of contract workers on call. Only activate them when the forecast shows demand crossing 228.

Recommendation 3: Negotiate spread-out appointment dates with Amazon The root cause of spikes is appointment clustering. Request Amazon to spread large order appointments across the month — minimum 5 days apart. This alone would reduce peak from 264 to approximately 150, cutting temp worker dependency by 60%.

Recommendation 4: Build a 7-day forward manpower forecast Using order data + appointment dates, automatically calculate workers needed for the next 7 days every Monday morning. Operations manager gets a weekly staffing plan instead of reacting to crises. This is a simple Python script — 30 lines of code — that can run automatically.